Unknown Breach: Evidence Discovery, Validation, and Full Forensic Investigation

Dataset: NIST CFReDS 2020JimmyWilson.E01

Tools Used: FTK Imager, Autopsy, Eric Zimmerman Registry Explorer, Velociraptor

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Executive Summary

This report presents a forensic investigation of the publicly available NIST CFReDS 2020JimmyWilson.E01 Windows forensic image. The investigation identified suspicious user-driven activity under the Jimmy Wilson profile, including pricing-related files, an encoded price list, BCTextEncoder.exe on the Desktop, suspicious browser artifacts related to hiding money, deleted files, and USB device activity. The evidence suggests possible concealment or obfuscation of financial/pricing-related data. However, no confirmed malware persistence, active malicious process, or direct proof of data exfiltration was identified

Dataset Selection

The NIST CFReDS 2020JimmyWilson.E01 forensic image was selected because it is a publicly available Windows forensic dataset containing realistic user activity. It was suitable for this project because it allowed analysis of browser artifacts, deleted files, USB usage, Windows registry artifacts, Prefetch, ShimCache, and suspicious user files. A limitation of the dataset is that it is a training image, so the activity may be simulated rather than from a real-world incident and the image did not allow investigating persistence mechanisms and amcache

Evidence Validation

The forensic image was validated using FTK Imager. MD5 and SHA1 hashes were calculated and compared to verify the integrity of the evidence. The hash values matched, confirming that the image was not altered before analysis

Image Name: 2020JimmyWilson.E01

Image Format: E01

MD5: b267fb0cd94645425eee00258d3a9b58

SHA1: a1102c70a50768b588225fdcad6efa5d5d57341b

Verification Result: Matched / Verified

Methodology

The investigation followed a structured forensic workflow. FTK Imager was used to validate, inspect, and mount the forensic image. Autopsy was used for disk analysis, including recent documents, Recycle Bin artifacts, browser artifacts, USB device usage, and keyword searches. Eric Zimmerman Registry Explorer was used to analyze registry artifacts such as Run/RunOnce keys, UserAssist, and ShimCache/AppCompatCache. Velociraptor was used on the mounted forensic image to simulate threat hunting by searching for known indicators of compromise, checking Startup folder persistence, and reviewing execution traces

Initial Triage

Initial triage confirmed that the image contained a Windows operating system based on the presence of the Windows directory and System32 files. Multiple user accounts were identified, including BillyBob, Fred Flintstone, James Russell, Jimmy Wilson, and Sgt. Joe T. Nameless. Jimmy Wilson became the main user of interest because suspicious pricing-related files, encoded files, and BCTextEncoder.exe were found under his user profile

Deep Disk Forensics

Recent document and LNK artifact analysis revealed references to pricing-related files, including NewPrices.txt, New Price List.lnk, and New Prices.lnk. These shortcut artifacts helped identify files and locations the user interacted with, supporting reconstruction of user activity even if the original files were moved or deleted

Recycle Bin analysis identified deleted user files, including a New Prices folder from the Desktop, a database file (.mdb), and text documents such as After All.txt. These files were associated with user activity and were removed after use, suggesting possible cleanup of potentially relevant information

USB artifact analysis identified multiple external devices connected to the system, including Lexar JumpDrive Secure, SanDisk Card Reader, and Kingston DataTraveler devices. These artifacts confirm removable media usage, which may represent a possible data transfer path, although no direct proof of file copying to USB was found

Keyword searches revealed repeated references to pricing-related files such as NewPrices.txt and related shortcut files on the Desktop. Searches also identified browser artifacts referencing “How to Hide Money From the Government,” indicating user interest in concealing financial assets or information

Windows Artifact Analysis

Event log analysis was performed to examine authentication and system activity. The available logs indicated normal system behavior, with no clear evidence of suspicious login attempts or abnormal authentication activity

Registry Run and RunOnce keys were examined using Eric Zimmerman Registry Explorer under Software\Microsoft\Windows\CurrentVersion\Run and RunOnce. No suspicious or unauthorized entries were identified, indicating no evidence of user-logon persistence through these keys

UserAssist registry entries were analyzed using Eric Zimmerman Registry Explorer to identify executed programs. The analysis showed normal user activity, including applications such as Snipping Tool, Paint, Sticky Notes, Media Center, and Solitaire. No clearly suspicious or malicious program execution was identified from UserAssist

Prefetch files were obtained from the Windows\Prefetch directory and analyzed to identify executed programs. The analysis mainly showed standard Windows processes such as DLLHOST.EXE, RUNDLL32.EXE, WMIADAP.EXE, WMIPRVSE.EXE, and WUAUCLT.EXE. No clear malicious execution trace was confirmed from Prefetch

ShimCache/AppCompatCache analysis was performed using Eric Zimmerman Registry Explorer by examining the SYSTEM hive under ControlSet001\Control\Session Manager\AppCompatCache. The analysis identified BCTextEncoder.exe in Jimmy Wilson’s Desktop path. This was relevant because an encoded price list was also found in the user’s files, suggesting possible use of an encoding tool to obscure pricing-related data

Amcache analysis was attempted by searching for Amcache.hve and manually checking the Windows\AppCompat\Programs directory. The Amcache.hve artifact was not present in the dataset, which limited executable metadata analysis through Amcache

Threat Hunting Simulation

Velociraptor was used to perform a threat-hunting simulation on the mounted forensic image. The hunt focused on known indicators discovered during previous analysis, especially BCTextEncoder.exe. Velociraptor identified BCTextEncoder.exe in G:\USERS\Jimmy Wilson\Desktop\, confirming the presence of a suspicious encoding tool linked to the encoded price list evidence

Velociraptor was also used to check Startup folder persistence locations. The only item identified in Jimmy Wilson's Startup folder was desktop.ini, which is normal Windows folder metadata. A Prefetch-focused search identified RUNDLL32.EXE, which is a common Windows process and was not suspicious by itself. No active malicious process or persistence mechanism was confirmed

Attack Reconstruction

The reconstructed incident suggests suspicious insider or user-driven activity rather than a confirmed external malware compromise. Jimmy Wilson appears to have interacted with pricing-related files, including a normal price list and an encoded version. Browser artifacts related to hiding money suggest possible financial concealment intent. BCTextEncoder.exe was present on the Desktop and was linked to the encoded price list evidence through ShimCache and Velociraptor findings

Persistence mechanisms were not identified, and execution artifacts did not confirm malware activity. Overall, the incident is best reconstructed as possible user-driven concealment or obfuscation of pricing-related data rather than a persistent malware-based compromise

Timeline Creation

The timeline was created by correlating file system artifacts, Recycle Bin entries, browser artifacts, USB artifacts, registry evidence, ShimCache, Prefetch, and Velociraptor findings

Timeline Table 1: Background / Early Activity

Timeline Table 2: USB / Possible Data Movement

Timeline Table 3: Pricing / Concealment Activity

Timeline Table 4: User Interaction / Execution Evidence

The timeline shows suspicious user-driven activity centered on Jimmy Wilson's pricing-related files. Evidence links the New Prices folder, encoded price list, BCTextEncoder.exe, and browser artifacts about hiding money. Deleted files and USB device activity suggest possible cleanup and data movement, but no direct proof of exfiltration was confirmed. Persistence checks were clean, and no active malicious process was identified

Data Exfiltration Hypothesis

Data exfiltration could not be conclusively confirmed from the available evidence. However, several artifacts support a possible exfiltration hypothesis, including the presence of removable USB devices, pricing-related files under Jimmy Wilson's profile, an encoded version of the price list, and BCTextEncoder.exe on the Desktop. Deleted files in the Recycle Bin further suggest possible cleanup activity

Although these findings indicate that sensitive pricing-related data may have been prepared or moved, no direct evidence of file transfer to USB, network upload, cloud storage, email attachment, or external destination was identified. Therefore, exfiltration remains possible but unproven

Limitations

This investigation was limited by the nature of the dataset and available artifacts. The evidence source was a forensic image rather than a live endpoint, so live process activity could not be confirmed. Amcache.hve was not present, limiting executable metadata analysis. USB artifacts confirmed device connections but did not prove specific file transfers. ShimCache entries indicate file presence or possible execution traces but do not conclusively prove execution by themselves

Recommendations

Based on the findings, organizations should monitor access to sensitive pricing and financial files, restrict unauthorized encoding or encryption tools, and log file transfers to removable media. USB device control policies should be enforced to reduce the risk of unauthorized data movement. Monitoring should also include suspicious browser activity, unusual file deletion patterns, and execution of tools from user directories such as Desktop, Downloads, AppData, and Temp

Conclusion

The investigation found evidence of suspicious financial-related user activity involving pricing files, encoded data, suspicious browser searches, deleted files, and removable media usage. The strongest evidence was the link between New Price List Encoded.txt and BCTextEncoder.exe found on Jimmy Wilson's Desktop. However, no confirmed malware persistence, active malicious process, or direct exfiltration evidence was identified. The most supported conclusion is suspicious insider activity involving possible concealment or obfuscation of pricing-related information